

2026 Chemical Engineering Process design competition

Problem Statement

1. Background:

As a process engineer, you've been given a new project to enhance your plant's cumene process. Cumene (isopropyl benzene) is the primary intermediate for phenol/acetone manufacturing and is produced by alkylating benzene with propylene over solid-acid catalysts according to the following reaction: $C_6H_6 + CH_2CHCH_3 \rightarrow C_6H_5CH(CH_3)_2$.

Before making changes in the plant, you'd like to build a simulation model of the process to use for optimization and running what-if scenarios. Most equipment in the process is easily modeled using appropriate thermodynamic methods and first-principles (aka equation-based) simulation. However, the kinetics of the cumene reactor are not well understood.

This has historically blocked development of the simulation model. Recently, you've been hearing more about Artificial Intelligence and think that a combination of Machine Learning (ML) with first-principles simulation could help overcome this challenge.

To obtain training data for the ML model, your team recently collected detailed data from the process. Analytical transmitters were installed around the process to measure the density at various points, including at the outlet of the reactor. You were also given approval to ramp the process between a variety of operating conditions during this time. Unfortunately, it is too costly to operate the analytical transmitters in normal production, and the process must now return to a normal operating state to meet existing production targets.

Your task is to develop a machine learning model trained on this real-world data and integrate it into Process Simulation to achieve a high-fidelity model that simulates the plant behavior.

2. Process Description

Process Flow Diagrams (PFDs) for the cumene process are shown in Figure 1 and Figure 2 below. Benzene is combined with propene in a 5:1 ratio of benzene to propene. In HX-1005, the mixed feed is heated to 350°C using a DOWTHERM™ heat transfer utility. The hot feed stream then enters an adiabatic packed bed reactor (R-1100) to produce the cumene product. The reactor effluent is cooled to 90°C in HX-1102 before entering drum V-1104 to vent off light gases, which are used as fuel gas in other areas of the plant. The remaining benzene/cumene mixture is fed to a 13-stage distillation column (T-1200) where a 99 mol% cumene product is recovered in the bottom product stream. From the top of the column, benzene (containing ≤ 1 mol% cumene) is recycled back to the front of the process and combined with fresh benzene in feed tank T-1001.

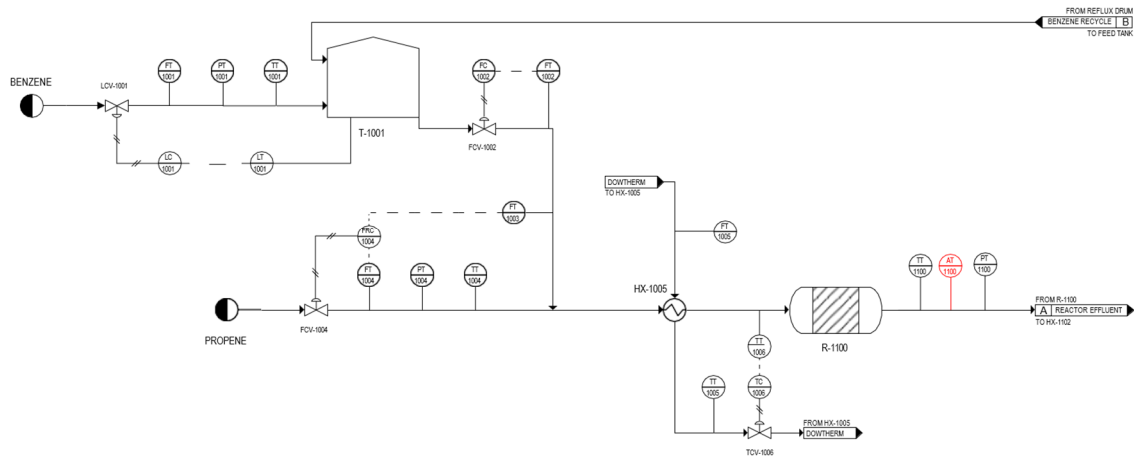


Figure 1 - Process Flow Diagram (PFD) for the reactor section of the cumene process

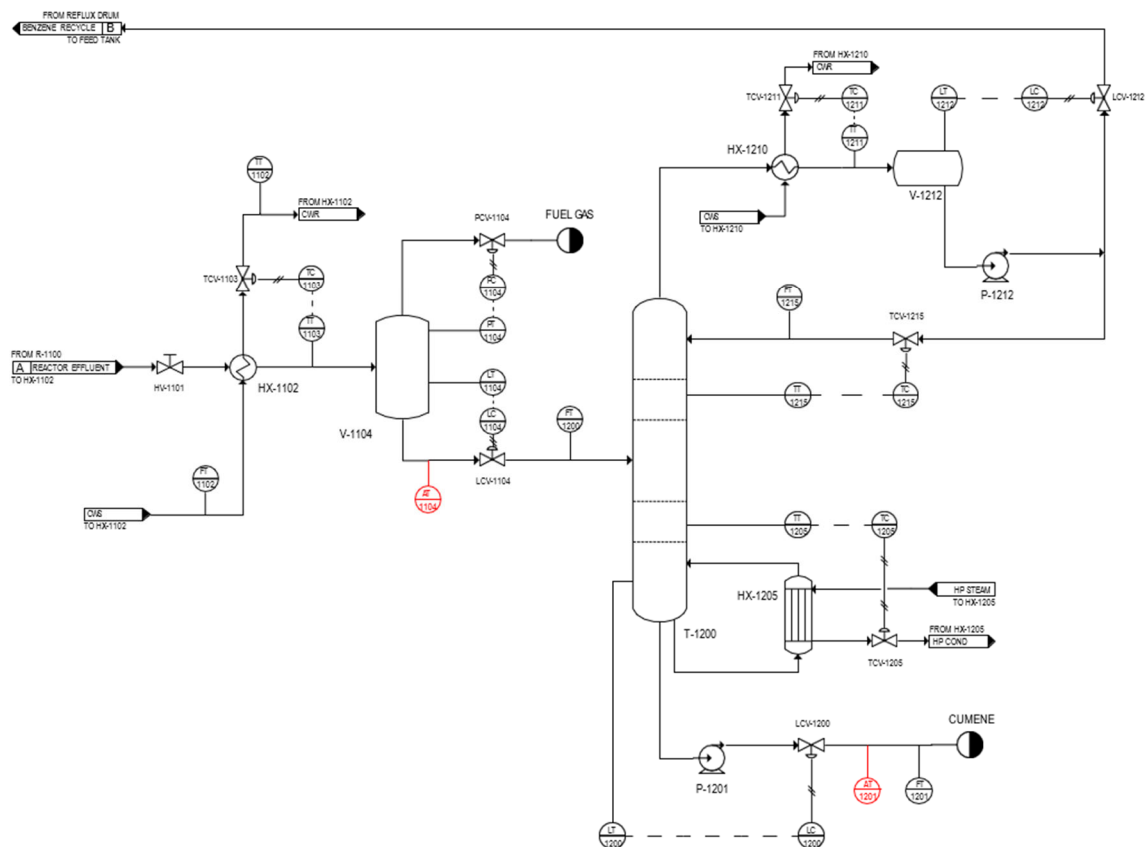


Figure 2 - Process Flow Diagram (PFD) for the separation section of the cumene process

3. Requirement

3.1 Machine Learning model development

You will use data (**attachment 1**) measured between October 12, 2025 and October 17, 2025 to train your ML model. Appendix A shows data list that is available to you. Your task is to create a ML model that can predict the product quality of the reactor outlet stream so that the expensive physical transmitters can be avoided.

Train an ML model to predict the performance of the cumene reactor based on current process conditions.

NOTE: *your Machine Learning model can be trained using the framework and algorithm of your choice. If you are just getting started, consider exploring scikit-learn in Python. There are numerous tutorials available online to help you get started. Whatever tool you choose, make sure your final model can be exported to ONNX format. This will be required when you use the model in Process Simulation. For a full list of supported frameworks, visit onnx.ai/supported-tools.*

HINT: *we have provided data from more transmitters than are required to train the model. Points will be awarded for selecting an appropriate subset of transmitters and justifying your decisions. When choosing your inputs and outputs, make sure to consider the problem you are trying to solve and the previous limitations that have prevented you from building a simulation model in the past.*

3.2 Build Simulation model

You will integrate your ML model into Process Simulation to create a high-fidelity model that simulates the plant's behavior. A simplified PFD for the process is shown in Figure 1 and 2. Using the ML model you trained in 3.1 :

Build a simulation model of the cumene process. Your simulation model should include all equipment shown in Figure 1 and 2, except any model(s) you have chosen to represent with your ML model. Additional information required to build the model is given in Appendix B – Simulation Inputs.

Integrate your trained ML model into the simulation. To help you get started, refer an example in attachment 2 - Incorporating an ONNX model into Process Simulation.

NOTE: *if a specific model specification is not provided in Appendix B – Simulation Inputs, use the default specification from Process Simulation. You can choose to update the specification based on design heuristics, but this is not mandatory.*

The entire process should be simulated, and the temperature, pressure, phase, and composition of each stream must be configured to be technically feasible. Recall that all other equipment in the process (heat exchangers, distillation column, flash drums, etc.) can be accurately modeled using first-principles simulation.

3.3 Deliverable

Prepare a report describing the Machine Learning model and process simulation model you have made. The report must be limited to a maximum of 20 pages, excluding references and supplementary.

The report should include:

1. A description of the model you chose for Machine Learning (gradient boosting, nearest neighbors, etc.) and an explanation of why you chose the model.
2. A list of the input and output variables of your ML model and a description of how you chose those variables.
3. A summary of the metrics and scoring used to evaluate the accuracy of your ML model.
4. A short description/discussion of the simulation model you have built. Include at least one screenshot showing an overview of your simulation model and heat and material balance.
5. A description/discussion of how you have integrated your ML model into the simulation. Be sure to describe the inputs and outputs of the ML model and how the model was integrated into the simulation. If you have replaced equipment items from the PFD with your ML model, be sure to describe why you replaced these models.

***HINT:** You were also asked to describe the inputs and outputs of your ML model. In the ML model part, you should focus on the available data and how that was used to train the model. In simulation part, focus on how these translate to specific input and output variables in the simulation.*

6. Case study results showing the impact of 1) reactor temperature, 2) feed flow rate, and 3) benzene to propene ratio on the cumene production. Use the following bounds for the case study:
 - Reactor temperature from 335 °C – 365 °C
 - Reactor feed flow rate from 15,000 – 18,000 kg/h
 - Benzene to propene ratio from 4 – 6

At a minimum, your case study results should include the molar concentration of cumene at the outlet of the reactor (if available through your model) and the mass flow rate of the cumene product. Other relevant information that describes the impact of these changes on the process will also be accepted. Be sure to include figures of the case study results, a discussion for each scenario, and whether you believe the results to be accurate (e.g., if you were an engineer, how would you convince others that your model is producing the correct results?).

Appendix A

Table 1 - Measurement tag descriptions

Tag Name	Units	Description
AT-1100.PV	kg/m3	Density of reactor outlet stream*
AT-1104.PV	kg/m3	Density of column feed stream*
AT-1201.PV	kg/m3	Density of cumene product stream*
FT-1001.PV	kg/h	Mass flow rate of fresh benzene
FT-1002.PV	kg/h	Mass flow rate of benzene (includes recycle stream)
FT-1003.PV	kg/h	Mass flow rate of benzene (includes recycle stream)
FT-1005.PV	kg/h	Mass flow rate of DOWTHERM™ heat transfer fluid supply
FT-1102.PV	m3/h	Volumetric flow rate of cooling water
FT-1104.PV	m3/h	Volumetric flow rate of vent gases
FT-1200.PV	kg/h	Mass flow rate of column feed
FT-1201.PV	kg/h	Mass flow rate of cumene product
FT-1215.PV	kg/h	Mass flow rate of column recycle stream
FT-1300.PV	kg/h	Mass flow rate of benzene recycle stream
PT-1001.PV	kPa	Pressure of fresh benzene feed stream
PT-1004.PV	kPa	Pressure of propene feed stream
PT-1100.PV	kPa	Pressure of reactor outlet stream
PT-1300.PV	kPa	Pressure of benzene recycle stream
TT-1001.PV	°C	Temperature of fresh benzene feed stream
TT-1004.PV	°C	Temperature of propene feed stream
TT-1005.PV	°C	Temperature of DOWTHERM™ heat transfer fluid return
TT-1006.PV	°C	Temperature of reactor inlet stream
TT-1100.PV	°C	Temperature of reactor outlet stream
TT-1102.PV	°C	Temperature of cooling water return stream
TT-1103.PV	°C	Temperature after product cooler
TT-1211.PV	°C	Temperature of column overhead stream
FT-1004.PV	kg/h	Mass flow rate of propene feed stream
FC-1002.SP	kg/h	Benzene mass flow rate controller set point
FC-1002.PV	kg/h	Benzene mass flow rate
FRC-1004.SP	-	Mass flow ratio controller set point
FRC-1004.PV	-	Mass flow ratio (Benzene / Propene)
LC-1001.SP	%	Benzene feed tank level controller set point
LC-1001.PV	%	Benzene feed tank level
LC-1104.SP	%	Flash drum level controller set point
LC-1104.PV	%	Flash drum level
LC-1200.SP	%	Column sump level controller set point

LC-1200.PV	%	Column sump level
LC-1212.SP	%	Reflux drum level controller set point
LC-1212.PV	%	Reflux drum level
PC-1104.SP	kPa	Flash drum pressure controller set point
PC-1104.PV	kPa	Flash drum pressure
TC-1006.SP	°C	Reactor inlet temperature controller set point
TC-1006.PV	°C	Reactor inlet temperature
TC-1103.SP	°C	Product cooler utility outlet temperature control set point
TC-1103.PV	°C	Product cooler utility outlet temperature
TC-1205.SP	°C	Column bottoms temperature controller set point
TC-1205.PV	°C	Column bottoms temperature (measured at stage 11)
TC-1211.SP	°C	Column overhead temperature controller set point
TC-1211.PV	°C	Column overhead temperature
TC-1215.SP	°C	Column top temperature controller set point
TC-1215.PV	°C	Column bottoms temperature (measured at stage 3)

*Measurement is no longer available in the plant

Appendix B

Table 1 – Thermodynamic properties for fluid

Variable / Parameter	Value
Thermo Type	Compositional
System	Soave-Redlich-Kwong (SRK)
Phases	Vapor/Liquid (VLE)

Table 2 - Components for fluid

Name	Full Name	Formula	Composition
BENZENE	Benzene	C6H6	0.63
PROPENE	Propylene	C3H6	0.34
PROPANE	Propane	C3H8	0.02
CUMENE	Cumene	C9H12	0.01

Table 3 – Source specifications

Model	Property	Variable	Value
PROPENE	Pressure	P	31 bar
	Temperature	T	210 °C
	Molar Composition	M[BENZENE]	0 kmol
		M[PROPENE]	0.95 kmol
		M[PROPANE]	0.05 kmol
		M[CUMENE]	0 kmol
BENZENE	Pressure	P	32 bar
	Temperature	T	30 °C
	Molar Composition	M[BENZENE]	1 kmol
		M[PROPENE]	0 kmol
		M[PROPANE]	0 kmol
		M[CUMENE]	0 kmol

Table 4 – Heat exchanger specifications

Model	Property	Variable	Value
HX-1005	Overall heat transfer coefficient	U	568 W/m ² -K
	Tube Outlet Temperature	Tto	350 °C
	Shell Outlet Temperature	Tso	350 °C
	Shell Side Pressure Drop	DPs	0.25 bar
	Tube Side Pressure Drop	DPT	0.25 bar
HX-1102	Overall heat transfer coefficient	U	450 W/m ² -K
	Tube Outlet Temperature	Tto	40 °C
	Shell Outlet Temperature	Tso	90 °C
	Shell Side Pressure Drop	DPs	0.25 bar
	Tube Side Pressure Drop	DPT	0.25 bar

Table 5 – Reactor specifications

Model	Property	Variable	Value
R-1100	Thermal operation	Operation	Adiabatic
	Reaction kinetics	RxnModelType	Unknown
	Reaction rate basis	Basis	CatalystMass
	Number of tubes	Nt	234
	Tube length	L	2 m
	Tube diameter	D	76.2 mm
	Catalyst density	rhoc	1600 kg/m ³

Table 6 – Column specifications

Model	Property	Variable	Value
T-1200	Number of ideal stages	NStages	13
	Feed stage	FeedStage	7
	Condenser option	Condenser	Internal
	Reboiler option	Reboiler	Internal
	Sump option	Sump	Single
	Top stage pressure	Ptop	1.75 bar
	Stage pressure drop	DPstage	0.01 bar
	Condenser pressure drop	DPcond	0.25 bar
	Reboiler pressure drop	DPreb	0.25 bar
	Condenser vapor fraction	VFcond	0 mol frac

Table 7 – Valve specifications

Model	Property	Variable	Value
HV-1101	Outlet pressure	P2	1.8 bar

Table 9 – Available utilities for process

Utility	Property	Variable	Value
Dowtherm	Pressure	P	5 bar
	Temperature	T	550 °C
Cooling Water	Pressure	P	5 bar
	Temperature	T	30 °C